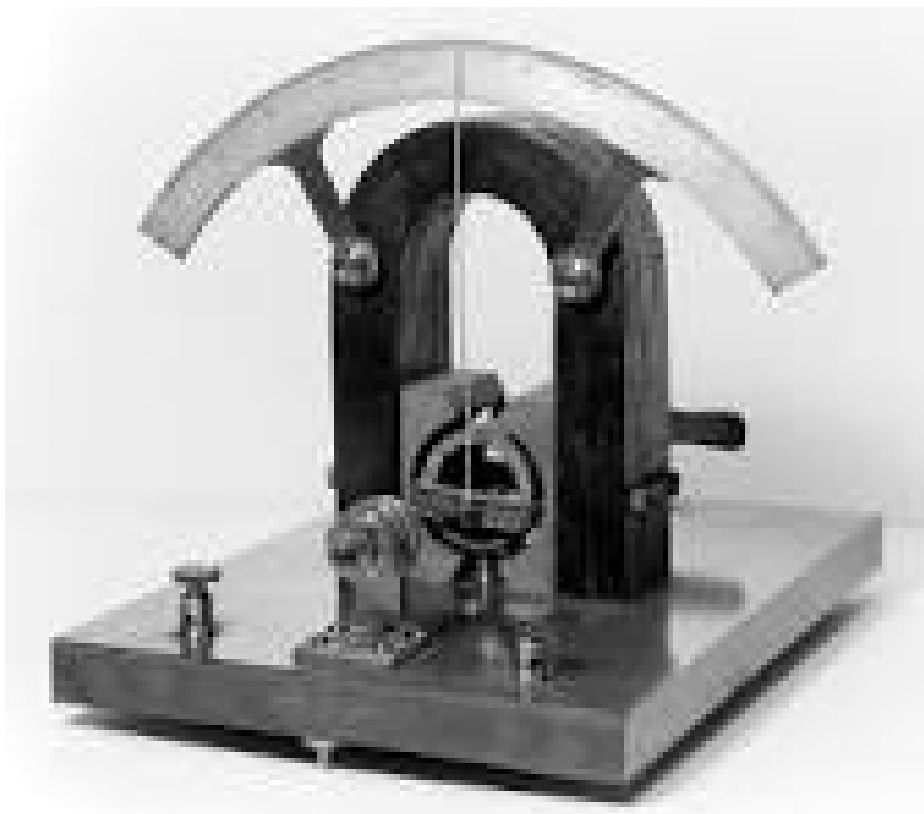


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Galvanometer



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Index

Introduction

A galvanometer is an instrument that measures electric current using a coil of insulated copper wire wound around a non-magnetic circular frame. Its operation relies on the tangent law of magnetism. When current passes through the coil, it generates a magnetic field (B) at the center of the coil, which is perpendicular to the plane of the coil.

The tangent galvanometer, an early type of current measuring device, utilizes a compass needle to compare the magnetic field created by an unknown current with the Earth's magnetic field. Its name comes from the tangent law of magnetism, which states that the tangent of the angle the compass needle makes is proportional to the ratio of the strengths of the two perpendicular magnetic fields. The circular scale of the galvanometer is divided into four quadrants, each marked from 0° to 90° .

To operate, the galvanometer is positioned with the plane of the coil vertical and aligned parallel to the horizontal component of the Earth's magnetic field (the local "magnetic meridian"). When electric current flows through the coil, it creates a second magnetic field that is perpendicular to the plane of the coil.

The magnetic field at the center of the coil is given by $B = \mu_0 n I / 2r$, where I is the current in amperes, n is the number of turns in the coil, and r is the radius of the coil. The two perpendicular magnetic fields combine vectorially, causing the compass needle to align with the resultant field $B_u + B$. The current in the coil can then be calculated using

$I = (2rBu/\mu n)\tan\theta$ where θ is the angle of deflection of the compass needle.

The current I can be expressed as $I = K \tan\theta$, where K represents the Reduction Factor of the tangent galvanometer. A limitation of the tangent galvanometer is that its accuracy decreases at both very high and very low current levels. The best resolution is achieved when the angle θ is 45 degrees. When θ is close to 0 or 90 degrees, even a significant change in current results in only a small movement of the needle.

Experiment

Aim:

To construct a simple galvanometer using only magnets and a wire and to find the extent of deflection for batteries with varying voltage.

Materials Required:

1. Plastic bottle
2. Thread
3. Two ring magnets
4. Cardboard
5. Copper wire
6. One AA cell
7. Pin
8. Blade
9. Tape

Procedure:

1. Cut a plastic bottle to create an open cylinder with a length of 7-10 cm.
2. Poke a hole in the bottle using a toothpick, through which you can thread a string.
3. Prepare a pointer and attach it to the end of a thread, then pass the thread through the hole.
4. Attach two ring magnets so that the pointer is positioned between them.
5. Wind copper wire around the cylinder in several loops.
6. Leave the ends of the wire exposed; these will serve as the positive and negative terminals of the galvanometer.
7. Cut a CD into an arc shape and mark it from 0 to 5A on either side of the central line.
8. Place the CD inside the bottle beneath the battery.
9. Secure the entire setup on a piece of cardboard with glue. Connect the wire ends to the positive and negative terminals of a AAA (1.5V), AA (3.6V), and 9V battery.
10. Record the deflections observed in the galvanometer in the observation table.

Observations

Result

The results from the pre-made galvanometer and the newly constructed one are consistent within the margin of experimental error. Therefore, the construction of the galvanometer has been successful.

Precautions

1. Ensure that the plastic bottle and CD fit together correctly for measuring deflection in the galvanometer.
2. Smooth the cut edges of the plastic bottle with sandpaper to prevent injury.
3. Make sure the wires are properly insulated and the coil is wound tightly.
4. Use a strong, twistable thread to suspend the magnet.

Applications

Galvanometers are utilized in:

1. Positioning and control systems
2. Laser engraving and laser sintering
3. Laser TVs and laser displays
4. Controlling the head mechanism
5. Positioning servos in hard drives and CD/DVD players
6. Obtaining photoresistor readings in the metering systems of film cameras

Bibliography

1. <https://youtu.be/ey2dlqwimd4> (link to youtube video used as reference in building the galvanometer)